Al-Balqa' Applied University
Faculty of Engineering Technology
Electronics1

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية

Midterm Exam
2024/2025

Time: 50 min.
23\11/2024

Student Name: [redacted] Student ID: [redacted]

(22 points)

Problem 1 Write the correct answers in the table in the last page.

1- **What types of impurity atoms are added to increase the number of conduction-band electrons in intrinsic silicon?**

a- bivalent b- octavalent c- pentavalent d) trivalent e- none of the above

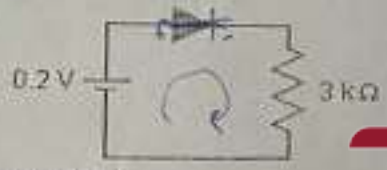
2- **What does a high resistance reading in both forward- and reverse-bias directions indicate?**

a. A good diode b) damage diode c) A short circuit diode d) A defective ohmmeter

3- **A silicon practical diode is forward-biased. You measure the voltage to ground from the anode at _____, and the voltage from the cathode to ground at _____.**

a-0 V, 0.3 V b-2.3 V, 1.6 V c-1.6 V, 2.3 V d- 0.3 V, 0 V

4- **Determine the voltage across the resistor 3 kΩ.**



a- 0.44V [illegible] c-0. [illegible] V

5- **To get a peak load voltage of 40 V out of a [illegible] What is the approximate rms value of secondary voltage?**

a) 0 V b) 14.4 V c) 28.3 V d) 56.6 V

6- **For a diode with $I_D = 25$ mA, the dynamic and DC resistance are .....**

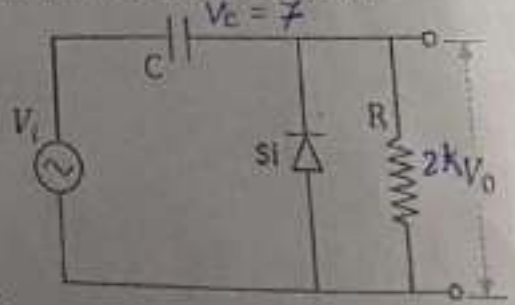
a. $R_{Dynamic} = 31.6\ \Omega$ and R_{DC} 31.6 Ω
b. $R_{Dynamic} = 31.6\ \Omega$ and R_{DC} 25 Ω
c. $R_{Dynamic} = 1.04\ \Omega$ and R_{DC} cannot be determined
d. $R_{Dynamic}$ cannot be determined and R_{DC} 1.04

$f_o = 2 f_{in}$

7- **In a full-wave rectifier circuit, if the output signal has a time period of 4 ms, then the input signal should have a frequency of ....**

a. 125 Hz b. 2 Hz c. 250 Hz d. 2 ms e. 250 KHz

8- **To implement the following circuit as a clamper for C = 7 μF and R = 2 KΩ with assuming a simplified diode is used, the discharge period of the input signal should be less than .....**



a. 14 b. 28 ms c. 7 ms d. 70 ms e. None of them.

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9-During the recombination process of *p-n* junction, the use of Ge diode will .....

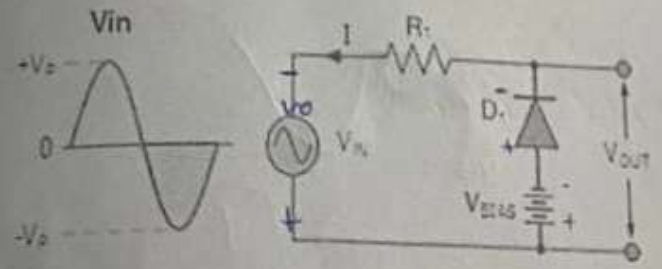
a-Emit a light in invisible zone

b-Emit light in the visible zone

c-Dissipate the energy as a heat

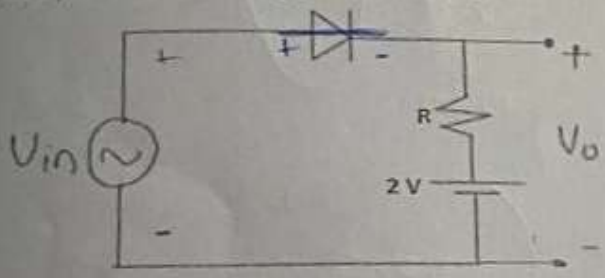
d-None of the above

10-The output voltage of the following circuit during the negative cycle is ..... (Assume a simplified model of Si)



a. – Vp b. + Vp c. VBias – 0.7 V d. -0.7 - VBias e. Vp -0.7 f. -Vp + VBias

11-When the diode for the following circuit is an ideal diode, the output voltage V_o when the diode is ON, R = 1 KΩ is equal to .....

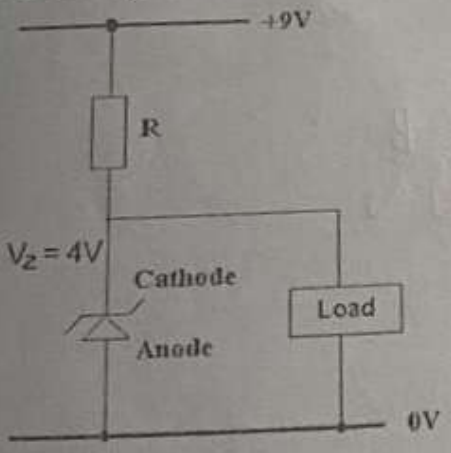


a. +2 V b. Vin c. – 0.7 VR f. None of them

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Problem 2 **(4 points)**

For the fig. below:



$I_{min} = 10mA$ $I_{max} = 90mA$

a-The minimum Zener diode current is 10 mA. If the maximum load current is 90 mA, calculat value for the series resistor R. $I_R = I_{Lmin} + I_{Z max} = 100mA$ (1)

$I_R = \frac{V_{in} - V_z}{R}$ $R = \frac{9}{100m} = 9\,\Omega$ (1.5)

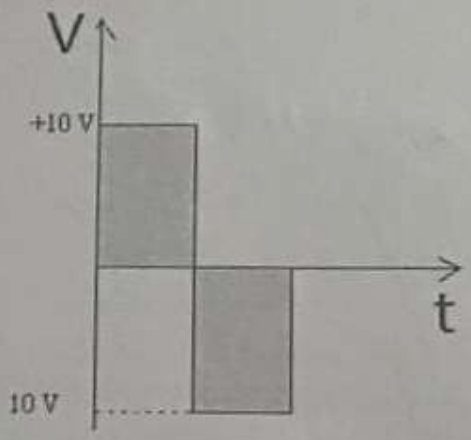
b-Calculate the power rating of the Zener diode in the diagram above.

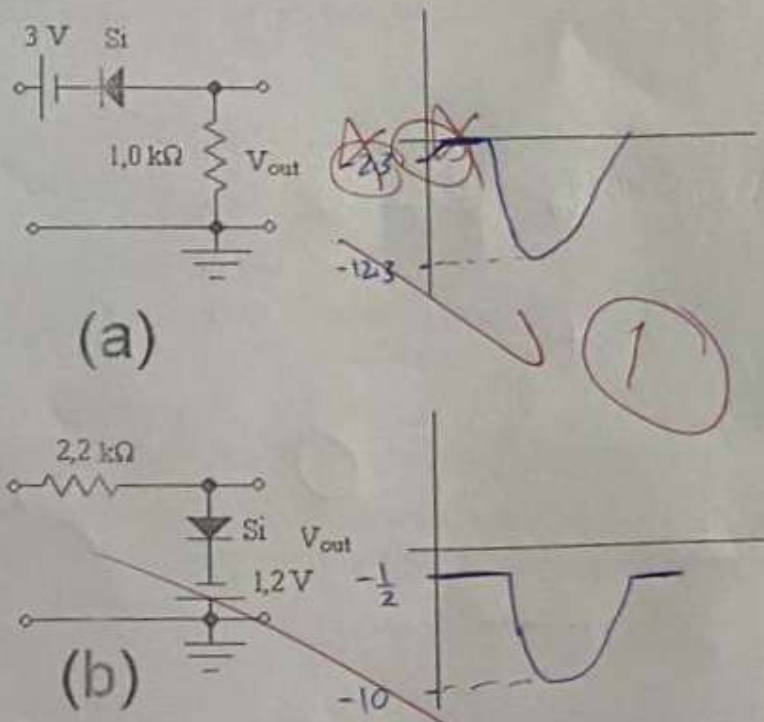
$P = IV$

$P = 0$

Poroblem 3 (4 points)

Problem 4: For the input shown, determine the output voltage waveform for each circuit in figure 3(a, b, c), the diodes in this circuit practical.





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Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	B	B	C	C	D	A	D	C	D	B

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Good Luck